Leonard N Jones II

Computational biologist with background in phylogenetics and spatial population genetics. Skilled in bioinformatics, data analysis and visualization, and communicating complex topics to broad audiences. Thrives in collaborative and creative environments.

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TECHNICAL SKILLS

Programming: R, python, SQL, Markdown, HTML

Environments: Unix/Linux, bash, Slurm scheduling, conda, Docker, github

Bioinformatics: SAMtools, GATK, BLAST, ClustalW, MAFFT, Bioconductor, tskit

Statistical analysis: Bayesian inference, clustering, regression, demographic modeling, geospatial analysis, PCA, ANOVA, t-test, SLiM

Wet lab: DNA and RNA extraction, Illumina NGS library preparation/processing, Sanger sequencing

PROFESSIONAL EXPERIENCE

Dept. of Ecology and Evolutionary Biology, University of Michigan — **Ann Arbor, MI**
NSF Postdoctoral Fellow, Bradburd Lab — **June 2022 - June 2024**

- Developed statistical models that account for both discrete and continuous spatial variation in admixture detection.
- Constructed an evolutionary simulation framework to test the efficacy of widely used geospatially agnostic methods for population structure inference under complex evolutionary histories.

Dept. of Integrative Biology, Michigan State University — **Lansing, MI**
Research Associate, Bradburd Lab — **Jan 2021 - May 2022**

- Conducted meta-analysis that identified perpetuation of bias in peer-review pipelines within Ecology and Evolution journals.

Dept. of Biology, University of Washington — **Seattle, WA**
Graduate Student Researcher, Leaché Lab — **Sept 2013 - Dec 2020**

- Conducted multi-year field research and specimen collection for deposition in museum collections, NGS library preparation, and data analysis for research on evolution in nonmodel systems. Communicated complex research topics to broad audiences across the knowledge spectrum, including the general public, community leaders, and industry stakeholders.
- Developed and taught lectures, discussions, and lab sections in introductory biology, evolution, advanced animal physiology, advanced cell biology, and evolutionary medicine.
- Secured over $100,000 in funding for teaching, research, and personal development initiatives from a diverse array of granting bodies.

Skirball Institute of Biomolecular Medicine **New York, NY**
Research Technician II, Lehmann Lab **Sept 2009 - May 2013**

- Conducted enhancer/suppressor genetic screen to identify candidate disruptors of germ cell development in *Drosophila melanogaster.*
- Developed cDNA library to examine conservation of spatial and temporal control mechanisms of germ-plasm RNA localization across *Drosophila* species.

PERSONAL DEVELOPMENT

Bonderman Travel Fellow **June 2024 - March 2025**
University of Washington **Multiple countries**

Awarded highly competitive fellowship to support eight months of independent international travel across nine countries in three distinct geographic regions.

Managed complex travel logistics and uncertainty, enhancing adaptability, problem-solving and operational planning in dynamic data-scarce environments with varying infrastructure limits.

Strengthened cross-cultural communication skills, frequently engaging with diverse communities and navigating language barriers to build meaningful connections and gather local insights.

SELECTED PUBLICATIONS

Jones LN *et al.* 2023. Biogeographic barriers and historic climate shape the phylogeography and demography of the common gartersnake. *Journal of Biogeography*, 50: 2012-2029. [**top 10% of most viewed publications in 2023**]

Theobald, EJ *et al.* 2020. Active learning narrows achievement gaps for underrepresented students in undergraduate science, technology, engineering, and math. *Proceedings of the National Academy of Sciences*, 117(12): 6476-6483. [**top 5% of Altmetric-scored research outputs**]

Leaché AD *et al.* 2015. Phylogenomics of Phrynosomatid lizards: conflicting signals from sequence capture versus restriction site associated DNA sequencing. *Genome Biology and Evolution*, 7(3): 706-719.

Hurd TH *et al.* 2013. Genetic modifier screens to identify components of a redox-regulated cell adhesion and migration pathway. *Methods in Enzymology*, 528: 197-215.

EDUCATION

University of Washington |PhD in Biology | Dec 2020
New York University |BA in Biology | Jan 2007

INTERESTS

natural history, cinema studies, cycling